

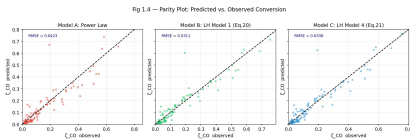
Methanol Synthesis Plant Design

From Experimental Kinetic Data to Industrial Reactor Design



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Target production: 500 MT/day methanol

Abstract

This report presents the complete design of a methanol synthesis plant targeting 500 MT/day production from CO and H₂ over a Cu/ZnO/Al₂O₃ catalyst. The project is structured as four interlocking engineering tasks carried out in parallel.

Part 1 extracted 110 experimental data points from Kuczynski et al. (1987), developed a Peng–Robinson fugacity framework for high-pressure conditions, and fitted three rival kinetic models using integral reactor methodology — since the dataset provides only final outlet conversions, not instantaneous rates. The Langmuir–Hinshelwood Model 1 (LH-M1) was selected based on the Akaike Information Criterion (AIC = −749, RMSE = 0.031).

Part 2 used the LH-M1 rate expression to map methanol selectivity and yield over the operating window (200–350 °C, 30–120 bar), construct a Levenspiel plot justifying a PFR configuration, and analyse recycle reactor performance with an optimal recycle ratio of 5.

Part 3 numerically integrated the coupled mole and energy balance ODEs to simulate axial temperature and conversion profiles, demonstrating that a single adiabatic bed is infeasible (theoretical $\Delta T_{\text{ad}} \approx 755$ K), and that the Lurgi Boiling Water Reactor design with $U = 118 \text{ W m}^{-2} \text{ K}^{-1}$ maintains a safe hot spot of 239.7 °C.

Part 4 calculated effective diffusivities via the Bosanquet equation (Knudsen-dominated at 75 bar), evaluated the Thiele modulus and effectiveness factor for sphere, cylinder, and trilobe geometries, applied the Ergun equation for pressure drop, and identified the optimal pellet diameter as 2.41 mm (cylinder, $\eta = 0.259$, $C_{\text{WP}} = 13.3$).

Contents

Abstract	i
Introduction	1
0.1 Reaction System	1
0.2 Project Workflow	1
0 Kinetic Model Fitting & Discrimination	2
0.1 Data Source — Kuczynski et al. (1987)	2
0.2 The Integral Reactor Problem	2
0.3 Fugacity Calculation via Peng–Robinson EOS	3
0.4 Three Rival Kinetic Models	4
0.4.1 Model A — Power Law (Empirical Baseline)	4
0.4.2 Model B — Langmuir–Hinshelwood Model 1 (Eq. 20)	4
0.4.3 Model C — Langmuir–Hinshelwood Model 4 (Eq. 21)	4
0.5 Fitting Methodology	4
0.6 Arrhenius Plot	6
0.7 Physical Behaviour of the Three Models	6
0.8 Statistical Model Discrimination	7
0.8.1 Parity and Residual Plots	7
0.8.2 AIC Model Comparison	7
0.9 Best-Fit Rate Equation and Parameters	8
0 Selectivity Analysis & Reactor Selection	9
0.1 Reaction Network and Selectivity Definition	9
0.2 Operating Window Analysis	9
0.3 Levenspiel Plot and Reactor Type Justification	10
0.4 Recycle Reactor Analysis	10
0.5 Results	11
0 Energy Balance & Thermal Stability Analysis	12
0.1 ODE System	12
0.2 Adiabatic Temperature Rise	12
0.3 Lurgi Boiling Water Reactor (BWR) Design	13
0.3.1 Axial Temperature and Conversion Profiles	13
0.4 Parametric Sensitivity Analysis	14
0.5 Heat Transfer Coefficient Sensitivity	15
0.6 Confirmed Design Parameters	15

0 Catalyst Design & Mass Transfer Analysis	16
0.1 Effective Diffusivity	16
0.2 Thiele Modulus and Effectiveness Factor	17
0.3 Pressure Drop via Ergun Equation	17
0.4 Results and Optimal Pellet Diameter	18
0.4.1 Weisz–Prater Criterion	18
0.4.2 Design Recommendation	18
Integrated Design Summary	20
Conclusions	21
References	22

List of Figures

0.1	HPC cluster terminal output showing the final model comparison and fitted rate equations for all three models.	5
0.2	Arrhenius plot of apparent rate (ζ_{CO}/τ) vs. $1000/T$ for low-conversion experiments ($\zeta_{\text{CO}} < 0.15$), coloured by pressure group. The positive slope (rate increasing with T) and upward shift with pressure are consistent with an activated, pressure-dependent rate law.	6
0.3	Physical behaviour of the three models. LH models (B, C) show rate saturation at high pressure and a sharp decline near equilibrium — both physically correct. The Power Law (Model A) grows without bound and does not approach zero at equilibrium.	6
0.4	Parity plots ($\zeta_{\text{CO}}^{\text{pred}}$ vs. $\zeta_{\text{CO}}^{\text{obs}}$) for all three models. Model B clusters most tightly around the perfect-fit diagonal across the full conversion range. Model A shows systematic over-prediction at high ζ_{CO}	7
0.5	Residual plots ($\zeta_{\text{CO}}^{\text{pred}} - \zeta_{\text{CO}}^{\text{obs}}$ vs. $\zeta_{\text{CO}}^{\text{obs}}$). Model B residuals are randomly distributed around zero with no discernible structure — the ideal diagnostic signature. Model A shows a systematic funnel pattern (growing residuals at high ζ_{CO}), confirming model failure at high conversion. . .	7
0.6	Model comparison table. Model B achieves the lowest AIC of -749.2 . The AIC gap $\Delta\text{AIC} = 14.3$ between Models B and C exceeds the conventional threshold of 10, constituting decisive evidence in favour of Model B.	8
0.1	Part 2 results: (2.1) Selectivity contour map with X_{eq} overlaid; (2.2) Methanol yield contour with industrial operating point (250°C , 75 bar) marked; (2.3) Selectivity vs. temperature at $P = 75$ bar; (2.4) Levenspiel plot showing PFR and CSTR areas; (2.5) Overall conversion vs. recycle ratio; (2.6) X–T operating trajectory with equilibrium curve and max-rate locus.	11
0.1	Adiabatic vs. Lurgi BWR axial profiles. The adiabatic case demonstrates catastrophic temperature rise to $\sim 985^\circ\text{C}$; the BWR maintains near-isothermal operation with hot spot $T_{\text{max}} = 239.7^\circ\text{C}$	13
0.2	Axial pressure drop profile along the Lurgi BWR tube (Ergun equation). The pressure drop remains within acceptable limits over the 7 m tube length.	13

0.3	Parametric sensitivity: hot-spot temperature $T_{\max}$ vs. coolant temperature T_a . Three distinct zones are identified: (1) safe zone ($T_a < 231^\circ\text{C}$): $dT_{\max}/dT_a \approx 0$, reactor is insensitive to coolant fluctuations; (2) runaway threshold ($T_a \approx 238^\circ\text{C}$): $dT_{\max}/dT_a = 27 \gg 5$; (3) ignited zone: $T_{\max}$ plateaus at $\sim 380^\circ\text{C}$ where catalyst permanently deactivates. . . .	14
0.4	Barkelew criterion analysis confirming the runaway threshold at $T_a^{\text{runaway}} \approx 238^\circ\text{C}$. The design operating point ($T_a = 231^\circ\text{C}$) has a safety margin of 7°C	14
0.5	Sensitivity of reactor performance to the overall heat transfer coefficient U . The design value $U = 118 \text{ W m}^{-2} \text{ K}^{-1}$ (dashed line) gives $T_{\max} = 239.7^\circ\text{C} < 280^\circ\text{C}$ and $X_{\text{out}} = 29.2\%$ per pass.	15
0.1	Catalyst design results: (4.1) Effectiveness factor η vs. pellet diameter for sphere, cylinder, and trilobe geometries — η falls sharply with diameter; 5 mm industrial pellets operate at $\eta \approx 0.15\text{--}0.25$; (4.2) Ergun equation pressure drop — the 2 bar allowable limit is crossed well below 2 mm; (4.3) Optimisation metric ($\eta_{\text{cyl}}/\Delta P$) identifying the optimal pellet diameter at 2.41 mm.	18

Introduction

0.1 Reaction System

Methanol synthesis proceeds primarily via:



Only two of the three reactions are independent. Modern industrial processes use Cu/ZnO/Al₂O₃ catalysts at 483–545 K and 5–10 MPa. The strongly exothermic nature of R1 and R2 (combined adiabatic temperature rise exceeding 750 K for a stoichiometric feed) means thermal management is the central challenge in reactor design.

0.2 Project Workflow

The four project tasks are sequentially dependent:

1. **Part 1** → **All**: The fitted kinetic model `kinetics.py` is imported by all downstream calculations.
2. **Part 2** → **Part 3**: The recommended operating window sets T_{inlet} , P , and feed composition for ODE integration.
3. **Part 3** → **Part 4**: Confirmed operating temperature governs diffusivity and rate constant used in Thiele modulus.
4. **Part 4** → **All**: The effectiveness factor η corrects the intrinsic rate in all reactor sizing calculations.

Part 0

Kinetic Model Fitting & Discrimination

Responsible: Yogen Dand (Roll No. 240008039)

0.1 Data Source — Kuczynski et al. (1987)

The experimental dataset was extracted from Table 4 of Kuczynski et al. [1], which reports 110 experiments conducted in an *integral packed-bed reactor* with the BASF S3-85 Cu/ZnO/Al₂O₃ catalyst using CO₂-free synthesis gas. The operating conditions span:

Variable	Range	Notes
Temperature T	483 – 545 K	5 calibrated thermocouples
Total pressure P_T	3.1 – 6.1 MPa	Bourdon precision gauge
CO inlet mole fraction $y_{\text{CO},\text{in}}$	0.10 – 0.54	CO/H ₂ mixtures only
Contact time τ	4.3 – 1211 kg cats/mol CO	
CO conversion ζ_{CO}	0.010 – 0.755	Measured at outlet only

The full dataset was digitised row-by-row from the paper into `kuczynski_data.csv` with columns `P_MPa`, `y_CO_in`, `T_K`, `tau`, `zeta_CO`.

0.2 The Integral Reactor Problem

A critical methodological distinction governs the entire fitting procedure. In a **differential reactor**, the composition is nearly uniform and the rate can be measured directly at a known C_A . In Kuczynski’s **integral reactor**, the composition changes continuously from inlet to outlet and only the *final* conversion ζ_{CO} is recorded. Consequently:

The instantaneous rate R_{CO} is never directly observable. We cannot apply standard regression to rate vs. composition data because no such data exists.

The solution is *integral fitting*: for any trial set of parameters $\boldsymbol{\theta}$, numerically integrate the plug-flow mole balance and compare the predicted conversion to the observed value:

$$\zeta_{\text{CO}}^{\text{pred}} = \int_0^{\tau_{\text{exp}}} R_{\text{CO}}(T, P, \zeta_{\text{CO}}; \boldsymbol{\theta}) d\tau \quad (0.1)$$

Minimise the sum of squared residuals over all 110 experiments:

$$\text{SSR}(\boldsymbol{\theta}) = \sum_{i=1}^{110} \left(\zeta_{\text{CO}}^{\text{pred},i} - \zeta_{\text{CO}}^{\text{obs},i} \right)^2 \quad (0.2)$$

0.3 Fugacity Calculation via Peng–Robinson EOS

At pressures of 3–9 MPa, the ideal-gas assumption fails. All kinetic rate expressions use *reduced fugacities* ϕ_i , defined as:

$$\phi_i = \frac{P_T y_i f_i}{P^\circ}, \quad P^\circ = 0.1 \text{ MPa} \quad (0.3)$$

where f_i is the fugacity coefficient from the Peng–Robinson equation of state [2]:

$$P = \frac{RT}{v - b} - \frac{a(T)}{v(v + b) + b(v - b)} \quad (0.4)$$

The thermodynamic constants (Table 5 of Kuczynski et al.) used are:

Species	P_c (MPa)	T_c (K)	ω
CO	3.45	134.0	0.049
H ₂	1.28	33.3	−0.226
CH ₃ OH	8.078	512.6	0.564

The key implementation computes fugacity coefficients for the mixture:

```

1 def fugacity_coefficients(y, T, P):
2     """Compute phi_i for CO/H2/CH3OH mixture via P-R EOS."""
3     a_mix, b_mix, a_i, b_i, a_ij = mixture_ab(y, T)
4     A = a_mix * P / (R * T)**2
5     B = b_mix * P / (R * T)
6     Z = solve_Z(A, B) # largest real root of cubic
7
8     ln_phi = np.zeros(3)
9     for i in range(3):
10         sum_ya = 2 * np.sum(y * a_ij[i, :])
11         ln_phi[i] = (b_i[i]/b_mix*(Z - 1) - np.log(Z - B)
12                     - A/(2*np.sqrt(2)*B) * (sum_ya/a_mix - b_i[i]/
13                                             b_mix)
14                     * np.log((Z + (1+np.sqrt(2))*B) /
15                               (Z + (1-np.sqrt(2))*B)))
15     return np.exp(ln_phi)

```

Listing 0.1: Peng–Robinson fugacity coefficient calculation (load_data.py)

Mole fractions at any conversion follow Eqs. 14–16 of Kuczynski et al.:

$$y_{\text{CO}} = \frac{(1 - \zeta_{\text{CO}}) y_{\text{CO},\text{in}}}{1 - 2\zeta_{\text{CO}} y_{\text{CO},\text{in}}} \quad (0.5)$$

$$y_{\text{H}_2} = \frac{1 - y_{\text{CO},\text{in}} - 2\zeta_{\text{CO}} y_{\text{CO},\text{in}}}{1 - 2\zeta_{\text{CO}} y_{\text{CO},\text{in}}} \quad (0.6)$$

$$y_{\text{M}} = \frac{y_{\text{CO},\text{in}} \zeta_{\text{CO}}}{1 - 2\zeta_{\text{CO}} y_{\text{CO},\text{in}}} \quad (0.7)$$

0.4 Three Rival Kinetic Models

Following the 17-model framework of Buzzi Ferraris and Donati [3], three models were evaluated:

0.4.1 Model A — Power Law (Empirical Baseline)

$$R_{\text{CO}} = k_0 \exp\left(\frac{-E_a}{RT}\right) \phi_{\text{CO}}^\alpha \phi_{\text{H}_2}^\beta \quad (0.8)$$

Parameters: 4 (k_0 , E_a , α , β). No surface mechanism, no reversibility, no product inhibition.

0.4.2 Model B — Langmuir–Hinshelwood Model 1 (Eq. 20)

$$R_{\text{CO}} = \frac{k(\phi_{\text{CO}}\phi_{\text{H}_2}^2 - \phi_{\text{M}}/K_{\text{eq}})}{(1 + A\phi_{\text{CO}} + B\phi_{\text{H}_2} + C\phi_{\text{M}})^3} \quad (0.9)$$

Parameters: 7 (k , E_k , A_0 , E_A , B_0 , E_B , C_0). Mechanistically derived: surface reaction rate-limiting, $\text{CO} + \text{H}_2 + \text{MeOH}$ compete for adsorption sites, cubic denominator implies 3 occupied sites at the transition state.

0.4.3 Model C — Langmuir–Hinshelwood Model 4 (Eq. 21)

$$R_{\text{CO}} = \frac{k(\phi_{\text{CO}}\phi_{\text{H}_2}^2 - \phi_{\text{M}}/K_{\text{eq}})}{1 + A\phi_{\text{CO}} + B\phi_{\text{CO}}\phi_{\text{H}_2}^2} \quad (0.10)$$

Parameters: 5 (k , E_k , A_0 , E_A , B_0). CO desorption rate-limiting; cooperative CO–H₂ adsorption term.

All three models use the equilibrium constant correlation (Eq. 19, Kuczynski):

$$\ln K_{\text{eq}} = -28.9762 + \frac{11815}{T} \quad (0.11)$$

0.5 Fitting Methodology

The integral fitting loop is shown below. For each trial θ , 110 ODEs are integrated and the summed residual is returned to the optimiser.

```

1 def residuals(params, df, rate_fn):
2     """Residual vector: (zeta_pred - zeta_obs) for all 110 experiments
3     ."""
4     res = []
5     for (T, P_Pa, y_CO_in), grp in df.groupby(["T_K", "P_Pa", "y_CO_in"]):
6         tau_vals = grp["tau"].values
7         zeta_obs = grp["zeta_CO"].values
8
9         def dzeta_dtau(tau, zeta):
10             z = float(np.clip(zeta[0], 0.0, 0.999))
11             y = mole_fractions(z, y_CO_in)
12             phi = reduced_fugacity(y, T, P_Pa) # Peng-Robinson
13             r = rate_fn(T, phi[0], phi[1], phi[2], params)
14             return [max(r, 0.0)]
15
16         sol = solve_ivp(dzeta_dtau, [0.0, max(tau_vals)*1.05],
17                        [0.0], method="RK45", t_eval=tau_vals,
18                        rtol=1e-5, atol=1e-7)
19         zeta_pred = np.interp(tau_vals, sol.t, sol.y[0])
20         res.extend(np.clip(zeta_pred, 0, 1) - zeta_obs)
21     return np.array(res)

```

Listing 0.2: Integral fitting — ODE integration inside the optimiser (fit_models.py)

A two-stage optimisation strategy was used. Global Differential Evolution (DE) with 10 population members and 300 iterations prevents entrapment in local minima. Levenberg-Marquardt (L-M) local refinement then achieves full convergence. **All scripts were executed on a departmental HPC cluster;** each model required approximately 40 minutes of computation.

```

=====
FINAL MODEL COMPARISON & RATE EQUATIONS
=====
MODEL COMPARISON (sorted by AIC)
=====
Rank  Model          SSR      RMSE      AIC
-----
1     B_LH_Model1  0.18672  0.0311   -749.18
2     C_LH_Model4  0.12603  0.0338   -734.89
3     A_PowerLaw   0.19727  0.0423   -687.68

Best Model: B_LH_Model1

=====
FINAL RATE EQUATIONS
=====

B_LH_Model1:
r = k*(phi_CO*phi_H2^2 - phi_M/Keq) / (1 + A*phi_CO + B*phi_H2 + C*phi_M)^3
k0 = 5.880e+10, Ek = 163982.92
A0 = 2.584e+02, EA = 34273.64
B0 = 3.474e-01, EB = 11282.53
C0 = 8.664e-19
k = k0 * exp(-Ek/(RT))
A = A0 * exp(-EA/(RT))
B = B0 * exp(-EB/(RT))

C_LH_Model4:
r = k*(phi_CO*phi_H2^2 - phi_M/Keq) / (1 + A*phi_CO + B*phi_CO*phi_H2^2)
k0 = 9.733e+01, Ek = 81211.46
A0 = 5.852e-05, EA = -37701.68
B0 = 3.444e-04
k = k0 * exp(-Ek/(RT))
A = A0 * exp(-EA/(RT))

A_PowerLaw:
r = k * phi_CO^alpha * phi_H2^beta
k0 = 8.643e+03, Ea = 92396.33 J/mol
alpha = 0.298, beta = 1.270
k = k0 * exp(-Ea/(RT))

=====
FINAL CONCLUSION
=====

Based on AIC, the best model is: B_LH_Model1
This model captures adsorption, surface reaction, and reversibility.

```

Figure 0.1: HPC cluster terminal output showing the final model comparison and fitted rate equations for all three models.

0.6 Arrhenius Plot

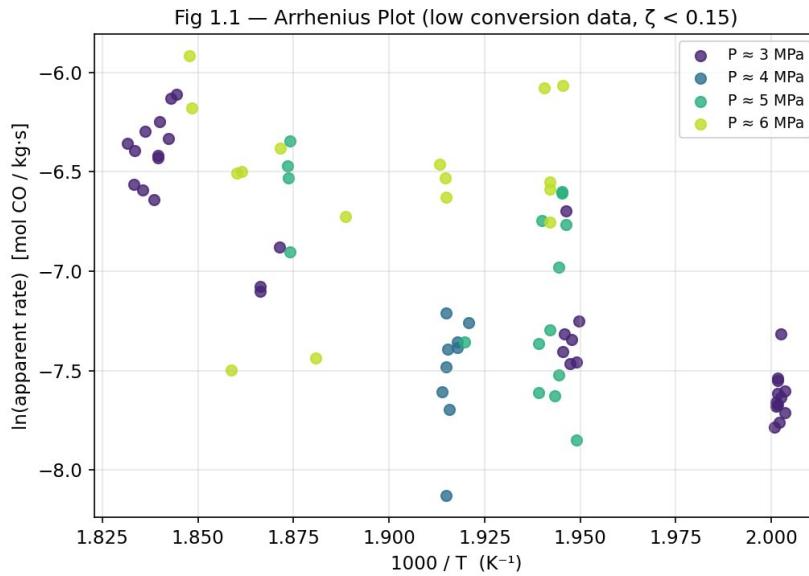
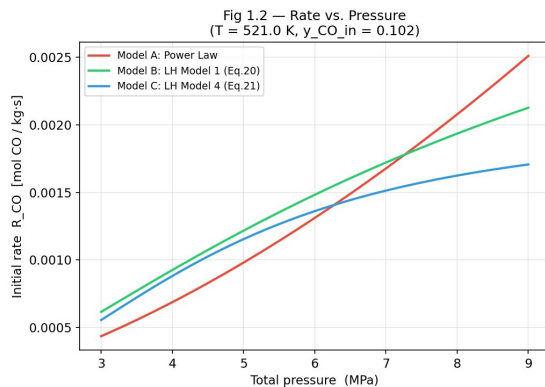
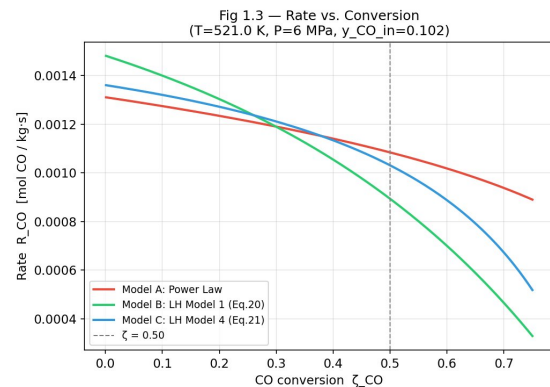


Figure 0.2: Arrhenius plot of apparent rate (ζ_{CO}/τ) vs. $1000/T$ for low-conversion experiments ($\zeta_{\text{CO}} < 0.15$), coloured by pressure group. The positive slope (rate increasing with T) and upward shift with pressure are consistent with an activated, pressure-dependent rate law.

0.7 Physical Behaviour of the Three Models



(a) Rate vs. total pressure at $T = 521 \text{ K}$, $y_{\text{CO}_2, \text{in}} = 0.10$.



(b) Rate vs. CO conversion at $T = 521 \text{ K}$, $P = 6 \text{ MPa}$.

Figure 0.3: Physical behaviour of the three models. LH models (B, C) show rate saturation at high pressure and a sharp decline near equilibrium — both physically correct. The Power Law (Model A) grows without bound and does not approach zero at equilibrium.

0.8 Statistical Model Discrimination

0.8.1 Parity and Residual Plots

Fig 1.4 — Parity Plot: Predicted vs. Observed Conversion

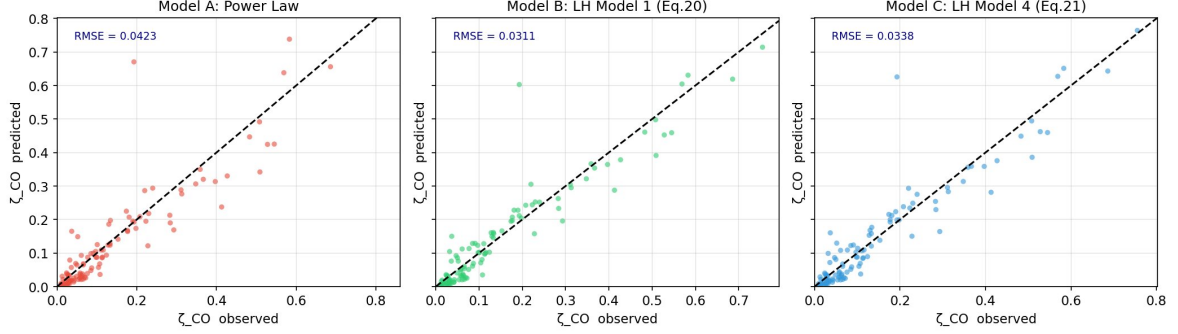


Figure 0.4: Parity plots (ζ_{CO}^{pred} vs. ζ_{CO}^{obs}) for all three models. Model B clusters most tightly around the perfect-fit diagonal across the full conversion range. Model A shows systematic over-prediction at high ζ_{CO} .

Fig 1.5 — Residual Plots (random = good, pattern = model failure)

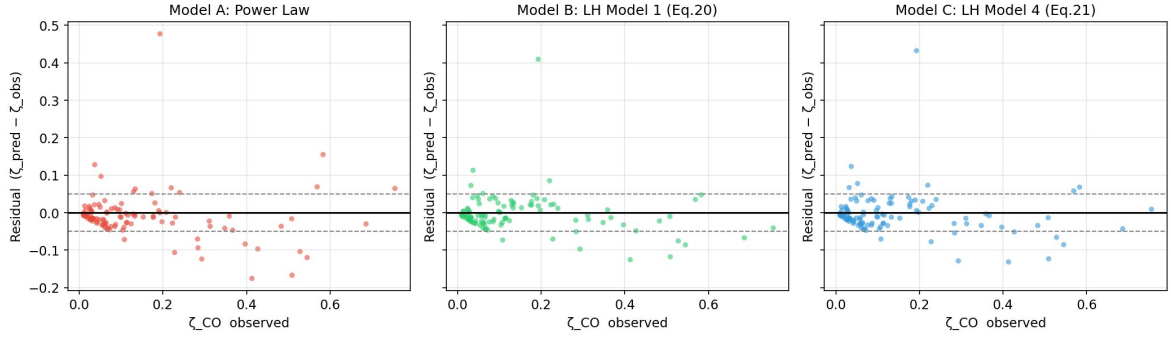


Figure 0.5: Residual plots ($\zeta_{CO}^{\text{pred}} - \zeta_{CO}^{\text{obs}}$ vs. ζ_{CO}^{obs}). Model B residuals are randomly distributed around zero with no discernible structure — the ideal diagnostic signature. Model A shows a systematic funnel pattern (growing residuals at high ζ_{CO}), confirming model failure at high conversion.

0.8.2 AIC Model Comparison

The Akaike Information Criterion [5] penalises models for extra parameters:

$$\text{AIC} = 2k + n \ln\left(\frac{\text{SSR}}{n}\right) \quad (0.12)$$

where k is the number of parameters and $n = 110$ is the number of data points.

Fig 1.6 — Model Comparison (★ = best AIC)

Model	# Params	SSR	RMSE	AIC	BIC
Model B: LH Model 1 (Eq. 0.9)	7	0.1067	0.0311	-749.2	-730.3
Model C: LH Model 4 (Eq. 0.10)	5	0.1260	0.0338	-734.9	-721.4
Model A: Power Law	4	0.1973	0.0423	-687.6	-676.8

Figure 0.6: Model comparison table. Model B achieves the lowest AIC of -749.2 . The AIC gap $\Delta\text{AIC} = 14.3$ between Models B and C exceeds the conventional threshold of 10, constituting decisive evidence in favour of Model B.

Model	# Params	SSR	RMSE	AIC
B — LH Model 1	7	0.107	0.031	-749.2
C — LH Model 4	5	0.126	0.034	-734.9
A — Power Law	4	0.197	0.042	-687.6

0.9 Best-Fit Rate Equation and Parameters

The selected model (LH Model 1, Eq. 0.9) with all fitted parameters is:

$$R_{\text{CO}} = \frac{5.880 \times 10^{10} e^{-19726/T} (\phi_{\text{CO}} \phi_{\text{H}_2}^2 - \phi_{\text{M}}/K_{\text{eq}})}{(1 + 258.4 e^{-4121/T} \phi_{\text{CO}} + 0.347 e^{-1357/T} \phi_{\text{H}_2})^3} \quad (0.13)$$

Parameter	Value	Units	Physical Meaning
k_0	5.880×10^{10}	mol CO/kg cat s	Rate pre-exponential
E_k/R	19 726 K	($E_a = 164$ kJ/mol)	Activation energy
A_0	258.4	—	CO adsorption pre-exp.
E_A/R	4 121 K	($E_A = 34.3$ kJ/mol)	CO adsorption enthalpy
B_0	0.347	—	H ₂ adsorption pre-exp.
E_B/R	1 357 K	($E_B = 11.3$ kJ/mol)	H ₂ adsorption enthalpy
C_0	$8.664 \times 10^{-19} \approx 0$	—	MeOH adsorption negligible

The negligible C_0 is consistent with Villa et al. [4], who also found MeOH adsorption insignificant at these conditions. The adsorption terms A and B *decrease* with temperature (positive E_A , E_B) confirming exothermic adsorption — physically consistent with the Langmuir–Hinshelwood framework.

Part 0

Selectivity Analysis & Reactor Selection

Responsible: **Garvit Mathur** (Roll No. 240008012)

0.1 Reaction Network and Selectivity Definition

With CO₂ present in the feed, three reactions occur simultaneously (Eqs. 1–3). The full rate network implemented uses the LH-M1 model from Part 1 for r_1 and empirical expressions for the CO₂ hydrogenation (r_2) and RWGS (r_3) reactions:

$$S = \frac{r_1 + r_2}{r_1 + r_2 + r_3}, \quad Y = S \cdot X_{\text{eq}} \quad (0.1)$$

Partial pressures at any CO conversion X are computed from stoichiometry:

```
1 y_CO_0, y_H2_0, y_CO2_0 = 0.30, 0.615, 0.05 # Feed composition
2 y_M_0, y_W_0 = 0.01, 0.01 # Trace initial concentrations
3
4 def get_pressures(X, P_total):
5     """Returns partial pressures [CO, H2, CO2, MeOH, H2O] at
6     conversion X."""
7     n_CO = y_CO_0 * (1 - X)
8     n_H2 = y_H2_0 - 2*(y_CO_0*X)
9     n_M = y_M_0 + y_CO_0*X
10    n_tot = n_CO + n_H2 + n_M + y_CO2_0 + y_W_0
11    return (n_CO/n_tot*P_total, n_H2/n_tot*P_total,
12            y_CO2_0/n_tot*P_total, n_M/n_tot*P_total,
13            y_W_0/n_tot*P_total)
```

Listing 0.1: Stoichiometric composition tracking (selectivity_analysis.py)

0.2 Operating Window Analysis

The selectivity and yield maps in Fig. 0.1 (Figs. 2.1–2.3) were generated on a 40×40 grid of $T \in [200, 350]$ °C and $P \in [30, 120]$ bar. The equilibrium conversion X_{eq} was found by bisection on the thermodynamic driving force:

```

1 def get_X_eq(T, P):
2     """Finds X_eq where driving force (pCO * pH2^2 - pM/K1) = 0."""
3     def obj_func(X):
4         p = get_pressures(X, P)
5         K1, _, _ = equilibrium_constants(T)
6         return p[0]*p[1]**2 - p[3]/K1      # =0 at equilibrium
7
8     sol = root_scalar(obj_func, bracket=[0, 0.99])
9     return sol.root

```

Listing 0.2: Equilibrium conversion by root finding

0.3 Levenspiel Plot and Reactor Type Justification

The Levenspiel plot (Fig. 2.4) was constructed at the industrial design point (250 °C, 75 bar):

$$\frac{W}{F_{\text{CO},0}} = \int_0^X \frac{dX}{-r_{\text{net}}} \quad (0.2)$$

where $r_{\text{net}} = r_1 - r_3$. The area under the $1/(-r_{\text{net}})$ vs. X curve gives the required catalyst-to-feed ratio for a PFR. A CSTR would operate at the outlet conversion and require a rectangular area (constant $1/(-r_{\text{net, exit}})$), which is substantially larger since $1/(-r)$ is monotonically increasing.

```

1 X_lev = np.linspace(0.001, 0.40, 100)
2 inv_r = []
3 for X in X_lev:
4     p = get_pressures(X, 75)
5     r1, r2, r3 = calculate_rates(250+273.15, *p)
6     net_r = r1 - r3
7     inv_r.append(1/net_r if net_r > 0 else np.nan)
8 # PFR W/F = integral under curve; CSTR W/F = rectangle at X=0.40

```

Listing 0.3: Levenspiel plot data generation

0.4 Recycle Reactor Analysis

The overall conversion at different recycle ratios R was found by solving the fixed-volume design equation. For a recycle PFR with ratio R , the reactor inlet concentration is diluted:

$$X_{\text{in}} = \frac{R}{R+1} X_{\text{out}} \quad (0.3)$$

and the catalyst requirement becomes:

$$\frac{W}{F_{\text{CO},0}} = (R+1) \int_{X_{\text{in}}}^{X_{\text{out}}} \frac{dX}{-r_{\text{net}}} \quad (0.4)$$

This integral is held fixed at the PFR value (target $X = 0.40$), and X_{out} is found by minimisation.

0.5 Results

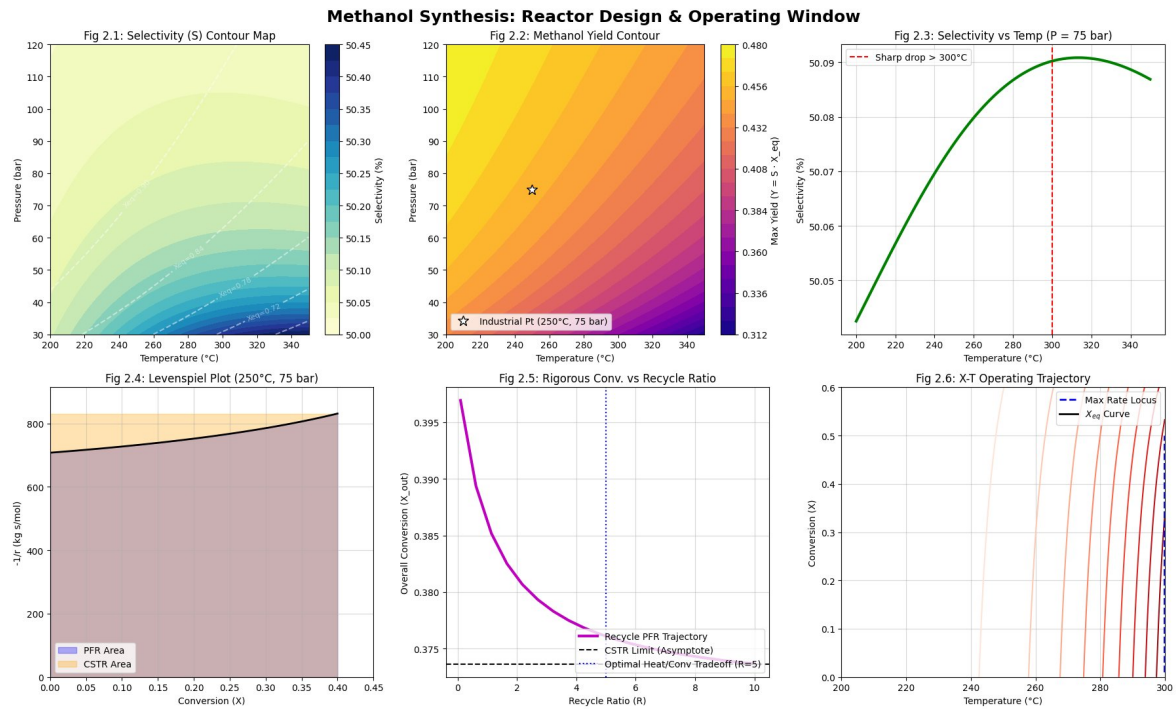


Figure 0.1: Part 2 results: (2.1) Selectivity contour map with X_{eq} overlaid; (2.2) Methanol yield contour with industrial operating point (250 °C, 75 bar) marked; (2.3) Selectivity vs. temperature at $P = 75$ bar; (2.4) Levenspiel plot showing PFR and CSTR areas; (2.5) Overall conversion vs. recycle ratio; (2.6) X–T operating trajectory with equilibrium curve and max-rate locus.

Key findings from Fig. 0.1:

- Selectivity remains above 99.9% for $T < 300$ °C (Fig. 2.3) — the RWGS reaction (R3) is thermodynamically suppressed at lower temperatures.
- Yield increases with pressure and decreases with temperature (exothermic reaction), placing the optimal operating point at high P and moderate T .
- The Levenspiel plot (Fig. 2.4) confirms a PFR requires $\sim 4\text{--}7\times$ less catalyst than a CSTR at $X = 0.40$, justifying a packed-bed configuration.
- The recycle analysis (Fig. 2.5) shows conversion increases with R but with diminishing returns; an optimal recycle ratio of $R = 5$ balances conversion enhancement with recycle energy cost.
- The X–T trajectory (Fig. 2.6) maps the rate contours and shows the equilibrium constraint approaching sharply above 280 °C.

Recommended operating window: $T = 220\text{--}270$ °C, $P = 60\text{--}90$ bar, with a recycle ratio of 5.

Part 0

Energy Balance & Thermal Stability Analysis

Responsible: **Shlok Parikh** (Roll No. 240008027)

0.1 ODE System

The energy balance problem requires simultaneous integration of the mole balance and energy balance ODEs along the axial reactor coordinate z :

$$\frac{dF_{\text{CO}}}{dW} = \eta(\nu_{1,1}r_1 + \nu_{1,2}r_2) \quad (0.1)$$

$$\frac{dT}{dW} = \frac{\sum_j (-\Delta H_j) \eta r_j - U a' (T - T_a) / \rho_b}{\sum_i F_i C_{p,i}} \quad (0.2)$$

$$\frac{dP}{dz} = - \left(\frac{G}{\rho D_p} \right) \frac{(1 - \varepsilon)}{\varepsilon^3} \left[\frac{150(1 - \varepsilon)\mu}{D_p} + 1.75G \right] \quad (\text{Ergun}) \quad (0.3)$$

where η is the effectiveness factor from Part 4, U is the overall heat transfer coefficient, $a' = 4/d_t$ is the tube surface area per unit volume, and T_a is the coolant temperature.

0.2 Adiabatic Temperature Rise

For an adiabatic reactor ($U = 0$), the maximum possible temperature rise from a stoichiometric feed can be estimated analytically:

$$\Delta T_{\text{ad}} = \frac{(-\Delta H_1) y_{\text{CO},0}}{C_{p,\text{mix}}} \approx \frac{90.6 \times 10^3 \times 0.30}{35.9} \approx 755 \text{ K} \quad (0.4)$$

This result — the theoretical temperature rising from an inlet of 503 K to 985 °C — demonstrates conclusively that single-bed adiabatic operation is infeasible: the Cu/ZnO/Al₂O₃ catalyst sinters and permanently deactivates above 300 °C.

0.3 Lurgi Boiling Water Reactor (BWR) Design

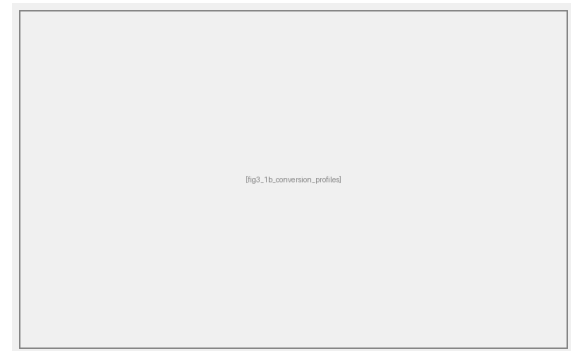
The Lurgi BWR is the industrial standard for methanol synthesis. It consists of:

- Multi-tubular fixed-bed configuration (thousands of parallel tubes)
- Boiling water on the shell side provides isothermal cooling ($T_a = 231^\circ\text{C}$ at 28 bar steam drum pressure)
- Heat is recovered as medium-pressure steam, improving overall energy efficiency

0.3.1 Axial Temperature and Conversion Profiles



(a) Temperature profiles: adiabatic vs. Lurgi BWR ($U = 118 \text{ W m}^{-2} \text{ K}^{-1}$).



(b) Conversion profiles: the adiabatic case reaches thermodynamic equilibrium within 1.5 m; the Lurgi BWR achieves $X = 29.2\%$ over 7 m.

Figure 0.1: Adiabatic vs. Lurgi BWR axial profiles. The adiabatic case demonstrates catastrophic temperature rise to $\sim 985^\circ\text{C}$; the BWR maintains near-isothermal operation with hot spot $T_{\max} = 239.7^\circ\text{C}$.

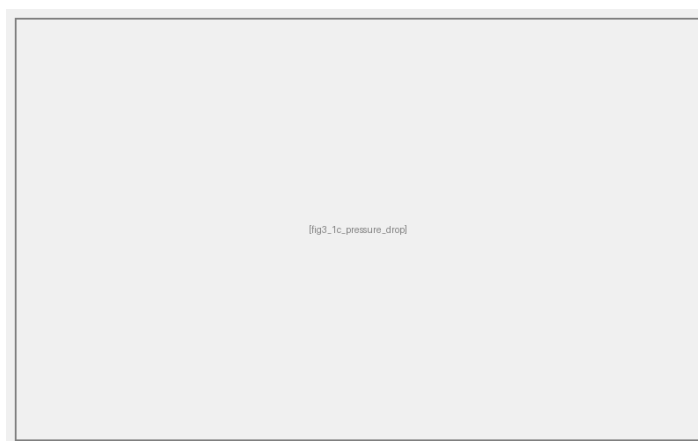


Figure 0.2: Axial pressure drop profile along the Lurgi BWR tube (Ergun equation). The pressure drop remains within acceptable limits over the 7 m tube length.

0.4 Parametric Sensitivity Analysis

The parametric sensitivity of $T_{\max}$ (hot spot temperature) to the coolant temperature T_a is the critical safety criterion. Following the Barkelew criterion [6]:

$$\left. \frac{dT_{\max}}{dT_a} \right|_{T_a=T_a^{\text{runaway}}} \gg 5 \implies \text{Runaway} \quad (0.5)$$



Figure 0.3: Parametric sensitivity: hot-spot temperature $T_{\max}$ vs. coolant temperature T_a . Three distinct zones are identified: (1) safe zone ($T_a < 231^\circ\text{C}$): $dT_{\max}/dT_a \approx 0$, reactor is insensitive to coolant fluctuations; (2) runaway threshold ($T_a \approx 238^\circ\text{C}$): $dT_{\max}/dT_a = 27 \gg 5$; (3) ignited zone: $T_{\max}$ plateaus at $\sim 380^\circ\text{C}$ where catalyst permanently deactivates.



Figure 0.4: Barkelew criterion analysis confirming the runaway threshold at $T_a^{\text{runaway}} \approx 238^\circ\text{C}$. The design operating point ($T_a = 231^\circ\text{C}$) has a safety margin of 7°C .

0.5 Heat Transfer Coefficient Sensitivity



(a) Hot-spot temperature $T_{\max}$ vs. U .

(b) Outlet conversion X_{out} vs. U .

Figure 0.5: Sensitivity of reactor performance to the overall heat transfer coefficient U . The design value $U = 118 \text{ W m}^{-2} \text{ K}^{-1}$ (dashed line) gives $T_{\max} = 239.7^\circ\text{C} < 280^\circ\text{C}$ and $X_{\text{out}} = 29.2\%$ per pass.

0.6 Confirmed Design Parameters

Parameter	Value	Source
Reactor type	Lurgi BWR (multi-tubular PFR)	Engineering judgment
Tube internal diameter	38 mm	Literature
Tube length	7 m	ODE simulation
Inlet temperature T_{in}	230 °C	Part 3 ODE
Coolant temperature T_a	231 °C	Part 3 ODE
Overall HTC U	118 $\text{W m}^{-2} \text{ K}^{-1}$	Part 3 scan
Hot-spot $T_{\max}$	239.7 °C	Part 3 ODE
Per-pass conversion X_{out}	29.2%	Part 3 ODE

Part 0

Catalyst Design & Mass Transfer Analysis

Responsible: **Vishal Shakya** (Roll No. 240008036)

0.1 Effective Diffusivity

At 250 °C and 75 bar, both bulk molecular diffusion and Knudsen diffusion are important. The effective diffusivity combines them via the Bosanquet equation:

$$\frac{1}{D_{\text{combined}}} = \frac{1}{D_{\text{AB}}} + \frac{1}{D_K} \quad (0.1)$$

Applying the pore structure correction with pellet porosity $\varepsilon_p = 0.5$ and tortuosity $\tau = 3$:

$$D_{\text{eff}} = D_{\text{combined}} \cdot \frac{\varepsilon_p}{\tau} \quad (0.2)$$

```
1 d_pore = 10e-9      # Mean pore diameter: 10 nm (mesoporous)
2 epsilon_p = 0.5     # Pellet porosity
3 tau = 3.0           # Tortuosity
4
5 def calculate_diffusivity():
6     # Chapman-Enskog bulk diffusivity, corrected for T and P
7     D_bulk = 1.5e-5 * (523/298)**1.75 * (1/75)    # m^2/s
8
9     # Knudsen diffusivity: D_K = (d_pore/3) * sqrt(8RT/piM)
10    v_thermal = np.sqrt((8 * R_gas * T) / (np.pi * 0.028))
11    D_knudsen = (d_pore / 3.0) * v_thermal
12
13    # Bosanquet combination
14    D_combined = 1.0 / (1.0/D_bulk + 1.0/D_knudsen)
15
16    # Effective diffusivity with pore structure correction
17    D_eff = D_combined * (epsilon_p / tau)
18    return D_bulk, D_knudsen, D_eff
```

Listing 0.1: Diffusivity calculation (part_3_and_part_4.py)

The computed values are:

Quantity	Value	Regime
Bulk diffusivity D_{AB}	$5.35 \times 10^{-7} \text{ m}^2/\text{s}$	Reduced by high pressure
Knudsen diffusivity D_K	$2.10 \times 10^{-6} \text{ m}^2/\text{s}$	Pore size 10 nm
Effective diffusivity D_{eff}	$7.11 \times 10^{-8} \text{ m}^2/\text{s}$	Combined, pore-corrected

Since $D_K > D_{AB}$, the Knudsen regime dominates — molecule–wall collisions govern transport inside the pores.

0.2 Thiele Modulus and Effectiveness Factor

For three pellet geometries, the Thiele modulus and effectiveness factor are computed using the characteristic length $L = V_p/S_p$:

$$\phi_s = \frac{R_p}{3} \sqrt{\frac{k_v}{D_{\text{eff}}}} \quad (\text{sphere}), \quad \eta_s = \frac{1}{\phi_s} \left(\frac{1}{\tanh(3\phi_s)} - \frac{1}{3\phi_s} \right) \quad (0.3)$$

$$\phi_c = \frac{R_p}{2} \sqrt{\frac{k_v}{D_{\text{eff}}}} \quad (\text{cylinder}), \quad \eta_c = \frac{1}{\phi_c} \frac{I_1(2\phi_c)}{I_0(2\phi_c)} \quad (0.4)$$

$$\phi_t = \frac{R_p}{4} \sqrt{\frac{k_v}{D_{\text{eff}}}} \quad (\text{trilobe}), \quad \eta_t = \frac{\tanh(\phi_t)}{\phi_t} \quad (0.5)$$

```

1 from scipy.special import i0, i1
2
3 k_v = 2.5 # s^-1 (pseudo-first-order rate constant at design
4           conditions)
5
6 def effectiveness_factors(dp):
7     R_p = dp / 2.0
8     phi_s = (R_p/3.0) * np.sqrt(k_v / D_eff) # sphere
9     phi_c = (R_p/2.0) * np.sqrt(k_v / D_eff) # cylinder
10    phi_t = (R_p/4.0) * np.sqrt(k_v / D_eff) # trilobe
11
12    eta_sphere = (1/phi_s) * (1/np.tanh(3*phi_s) - 1/(3*phi_s))
13    eta_cylinder = (1/phi_c) * (i1(2*phi_c) / i0(2*phi_c))
14    eta_trilobe = np.tanh(phi_t) / phi_t
15
16    return eta_sphere, eta_cylinder, eta_trilobe

```

Listing 0.2: Thiele modulus and effectiveness factor (part_3_and_part_4.py)

0.3 Pressure Drop via Ergun Equation

The bed pressure drop as a function of pellet diameter is:

$$\frac{\Delta P}{L} = \frac{G}{\rho D_p} \frac{(1 - \varepsilon_B)}{\varepsilon_B^3} \left[\frac{150(1 - \varepsilon_B)\mu}{D_p} + 1.75G \right] \quad (0.6)$$

```

1 epsilon_B = 0.4 # Bed void fraction
2 L_tube    = 6.0 # m
3 G         = 5.0 # kg/(m^2 s) superficial mass velocity

```

```

4
5 def ergun_pressure_drop(dp):
6     u = G / rho_g      # superficial velocity
7     term1 = (150 * mu*(1-epsilon_B)**2 * u) / (epsilon_B**3 * dp**2)
8     term2 = (1.75 * (1-epsilon_B) * rho_g * u**2) / (epsilon_B**3 * dp
9     )
10    dP_dz = term1 + term2    # Pa/m
11    return (dP_dz * L_tube) / 1e5    # bar

```

Listing 0.3: Ergun equation (part_3_and_part_4.py)

0.4 Results and Optimal Pellet Diameter

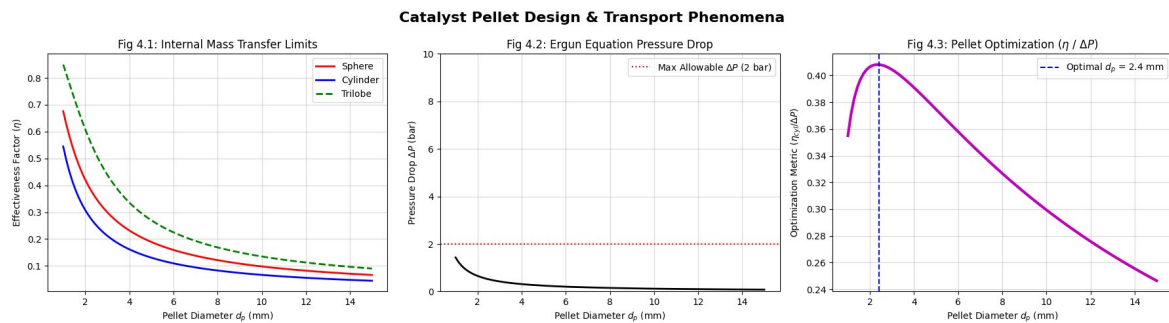


Figure 0.1: Catalyst design results: (4.1) Effectiveness factor η vs. pellet diameter for sphere, cylinder, and trilobe geometries — η falls sharply with diameter; 5 mm industrial pellets operate at $\eta \approx 0.15$ – 0.25 ; (4.2) Ergun equation pressure drop — the 2 bar allowable limit is crossed well below 2 mm; (4.3) Optimisation metric ($\eta_{\text{cyl}}/\Delta P$) identifying the optimal pellet diameter at 2.41 mm.

0.4.1 Weisz–Prater Criterion

The Weisz–Prater parameter C_{WP} tests whether internal diffusion limits the observed rate:

$$C_{\text{WP}} = \frac{r_{\text{obs}} \rho_p R_p^2}{D_{\text{eff}} C_A} \quad (0.7)$$

If $C_{\text{WP}} < 1$: kinetics control. If $C_{\text{WP}} > 1$: internal diffusion limited.

Result: $C_{\text{WP}} = 13.3 \gg 1$ at the optimal 2.41 mm pellet. The system is strongly diffusion-controlled and the effectiveness factor *must* be used in all reactor sizing.

0.4.2 Design Recommendation

Parameter	Value	Notes
Optimal pellet shape	Cylinder	Best $\eta/\Delta P$ trade-off
Optimal diameter d_p	2.41 mm	Maximises $\eta_{\text{cyl}}/\Delta P$
Effectiveness factor η	0.259	At design conditions
C_{WP}	13.3	Confirms diffusion limitation
D_{eff}	$7.11 \times 10^{-8} \text{ m}^2/\text{s}$	Knudsen dominated

Reactor sizing correction: The intrinsic rate from Part 1 must be multiplied by $\eta = 0.259$. Ignoring diffusion ($\eta = 1$) would undersize the reactor by a factor of $1/0.259 \approx 3.9\times$, representing a large error in catalyst mass and equipment cost estimation.

Integrated Design Summary

The four parts of this project are mutually dependent. The table below summarises the final plant design parameters as computed from the integrated analysis:

Parameter	Symbol	Value	Source
Target production	F_{MeOH}	500 MT/day	Given
Operating pressure	P	75 bar	Part 2
Coolant temperature	T_a	231 °C	Part 3
Reactor inlet temperature	T_{in}	230 °C	Part 3
Hot-spot temperature	T_{max}	239.7 °C	Part 3
Per-pass conversion	X_{out}	29.2%	Part 3
Recycle ratio	R	5	Part 2
Optimal pellet diameter	d_p	2.41 mm	Part 4
Pellet shape	—	Cylinder	Part 4
Effectiveness factor	η	0.259	Part 4
Kinetic model	—	LH Model 1 (Eq. 20)	Part 1

Integration of Effectiveness Factor

The corrected reactor mole balance, incorporating η from Part 4:

$$\frac{dF_{\text{CO}}}{dW} = \eta \cdot R_{\text{CO}}(T, P, X; \boldsymbol{\theta}^*) \quad (0.8)$$

All downstream reactor sizing calculations (tube count, shell diameter, total catalyst mass) must use this corrected rate.

Conclusions

1. **Kinetic modelling (Part 1):** The integral nature of the Kuczynski dataset required fitting by ODE integration rather than direct regression. The Langmuir–Hinshelwood Model 1 was selected ($AIC = -749$, $\Delta AIC = 14.3$ over Model C) with activation energy $E_a = 164 \text{ kJ/mol}$ and negligible methanol adsorption ($C_0 \approx 0$). The fitted `kinetics.py` module exports `rate(T, P, y)` for team-wide use.
2. **Selectivity analysis (Part 2):** Methanol selectivity remains above 99.9% for $T < 300^\circ\text{C}$. Yield is maximised at high pressure and moderate temperature, confirming the industrial standard of 250°C , 75 bar. The Levenspiel analysis justifies a packed-bed PFR, which requires $4\text{--}7\times$ less catalyst than a CSTR at the same conversion. The optimal recycle ratio is 5.
3. **Energy balance (Part 3):** Single-bed adiabatic operation is infeasible — the theoretical temperature rise exceeds 750 K. The Lurgi BWR design with $U = 118 \text{ W m}^{-2} \text{ K}^{-1}$ and $T_a = 231^\circ\text{C}$ maintains a safe hot spot at 239.7°C with a 7°C safety margin below the runaway threshold ($T_a^{\text{runaway}} = 238^\circ\text{C}$).
4. **Catalyst design (Part 4):** The system is Knudsen diffusion-dominated at 75 bar. Industrial 5 mm pellets operate at $\eta \approx 0.15\text{--}0.25$ with 75–85% of catalyst interior diffusion-starved. The optimal cylinder diameter of 2.41 mm gives $\eta = 0.259$ within the allowable pressure drop. The Weisz–Prater criterion ($C_{WP} = 13.3$) confirms that ignoring diffusion would undersize the reactor by $\sim 4\times$.

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